Sri Lanka Institute of Information Technology



IT3060 - Human-Computer Interaction

Milestone 1
User Research Techniques

Project Title – **Transport Tracking & Digital Ticketing App**

Assignment Group ID – Y3.S2.WE_62

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Group Member Details & Work Distribution

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Rajapaksha M.P.K	IT23665620	• Prepare the Interview Script for the Primary Stakeholder – Daily Bus/Train Commuter • Conduct interviews with the Primary Stakeholder • Create the Primary Persona – Daily Bus/Train Commuter • Create the Primary Empathy Map • Create the Primary User Journey Maps – Searching for a Bus/Train, Tracking the Vehicle, Purchasing a Digital Ticket, and Completing the Journey • Write User Stories 1 & 2 (Daily Passenger/Commuter) • Prepare the Gantt Chart
Herath H.M.S.G	IT23634626	• Prepare the Interview Script for the Secondary Stakeholder – Bus Conductor / Train Ticket Inspector • Conduct interviews with the Secondary Stakeholder • Create the Secondary Persona – Bus Conductor / Train Ticket Inspector • Create the Secondary Empathy Map • Create the Secondary User Journey Map – Validating Digital Tickets During a Journey • Write User Stories 3 & 4 (Conductor/Ticket Inspector)
Fernando H.L.R.D	IT23635302	• Prepare the Interview Script for the Tertiary Stakeholder – Transport Authority Operations Officer (NTC / Sri Lanka Railways) • Conduct interviews with the Tertiary Stakeholder • Create the Tertiary Persona – Transport Authority Operations Officer • Create the Tertiary Empathy Map • Write User Story 5 (Transport Authority) • Create the How Might We (HMW) Statements

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1 Project Title

RideTrack: A Real-Time Public Bus/Train Tracking and Digital Ticketing Application

2 Problem Statement

Public transportation is the main means of travel for many commuters in Sri Lanka, but passengers often lack a dependable method to find out when a bus or train will truly arrive. Schedules are frequently old or overlooked in reality, with delays, malfunctions, and traffic situations never relayed to passengers waiting. Consequently, riders often find themselves waiting endlessly at stops and stations, losing connections, or arriving late to work, classes, and appointments.

Ticketing introduces an additional level of difficulty. The majority of passengers continue to buy tickets in cash from a conductor or at a station counter, resulting in queues, hindering the boarding process, and complicating fare verification or accurate passenger count tracking for conductors. Bus and train operators, along with regulatory entities like the National Transport Commission (NTC) and Sri Lanka Railways, also do not have real-time insights into vehicle positions, delays, and occupancy levels, restricting their capability to manage the network effectively and react swiftly to disruptions.

RideTrack is suggested as a mobile app that displays the real-time location and anticipated arrival time of buses and trains, enables passengers to buy and verify digital tickets through QR codes, and equips conductors and transport authorities with the necessary tools to oversee journeys, confirm fares, and track the network live. The objective is to enhance the reliability of public transport, simplify payment processes, and streamline management for all parties concerned.

3 Problem Justification

Dependable and efficiently managed public transportation is crucial for a thriving city, but the unpredictability and hassle linked to existing bus and train services drive numerous commuters to private cars or three-wheelers, exacerbating traffic jams and pollution. Enhancing the passenger experience through improved information and simplified ticketing is one of the most effective methods to make public transport a truly appealing choice.

A mobile app is ideal since almost every commuter has a smartphone and can easily check it before departing home or while waiting at a stop. RideTrack can minimize idle waiting time, decrease cash-handling lines at boarding, and provide transport operators with the operational insight required to deliver a more dependable service by integrating live vehicle tracking, digital ticketing, and disruption alerts.

This initiative was chosen because it tackles an issue impacting a significant number of daily travelers, holds evident public value beyond a single user group, and presents a solid chance to implement Human-Computer Interaction (HCI) principles among three distinct user types: regular commuters, on-site transport staff, and transport authority employees in back-office roles.

4 Stakeholder Identification

4.1 Primary Stakeholders

These are the users who directly interact with the Real-Time Public Bus/Train Tracking and Digital Ticketing Application.

- **Daily commuters** – People who regularly use buses and trains to travel to work, university, school, or other destinations.
- **Students** – University and school students who depend on public transportation for their daily travel.
- **Occasional passengers** – Users who use public buses or trains occasionally for personal trips, appointments, shopping, or other activities.
- **Tourists and visitors** – People who may rely on public transportation to travel between unfamiliar locations.

Role: Search for available bus/train routes, view real-time vehicle locations and estimated arrival times, check schedules and fares, purchase digital tickets, and manage their journey information.

4.2 Secondary Stakeholders

These are the users who directly support and operate the public transport services through the Real-Time Public Bus/Train Tracking and Digital Ticketing Application.

- **Bus conductors –** Staff members responsible for checking passenger tickets, validating digital QR tickets, and assisting passengers during bus journeys.
- **Bus drivers –** Drivers responsible for operating buses and providing accurate vehicle location information through the tracking system.
- **Train ticket inspectors** – Railway staff responsible for checking and validating passenger tickets and preventing unauthorized travel.
- **Station staff –** Employees who assist passengers, manage station operations, and handle ticketing and service-related issues.

Role: Validate digital tickets using QR codes, view passenger and ticket information, monitor passenger counts, assist passengers, report operational issues, and support the smooth operation of bus and train services.

4.3 Tertiary Stakeholders

These are the organizations and personnel responsible for managing, monitoring, and maintaining the public transportation system and the application.

- **National Transport Commission (NTC) –** The regulatory authority responsible for overseeing and managing inter-provincial passenger bus transportation services.
- **Sri Lanka Railways Management –** Responsible for managing railway operations, train services, schedules, and overall railway service performance.
- **Bus company owners/operators –** Organizations or individuals responsible for operating buses, managing routes, monitoring vehicle performance, and maintaining service quality.
- **System administrators –** Technical personnel responsible for maintaining the application, managing users and system settings, monitoring system performance, and ensuring data security.

Role: Monitor bus and train operations, analyze delays and passenger statistics, manage routes and services, oversee transportation performance, generate operational reports, manage system users, and maintain the overall system.

5 User Research Plan

5.1 Research Objectives

The main objective of this research is to understand how commuters currently plan and complete public

transport journeys, identify the challenges they face with unpredictable schedules and cash-based ticketing, and

determine the features they expect from a real-time tracking and digital ticketing mobile application.

The specific objectives are:

- To understand how passengers currently find out when a bus or train will arrive.
- To identify common challenges in waiting for, boarding, and paying for public transport.
- To explore users' expectations for a live tracking and digital ticketing mobile application.
- To gather requirements from passengers, transport staff, and transport authority personnel to guide the design of RideTrack

5.2 Research Method

A survey utilizing a questionnaire was chosen as the main research approach to engage primary stakeholders (passengers) on a large scale. Google Forms was utilized to disseminate the survey since it facilitates the effective collection of responses from commuters with various travel habits and profiles. Semi-structured interviews were subsequently carried out with representative individuals from each of the three stakeholder groups — a regular commuter, a bus conductor/ticket inspector, and an operations officer from the transport authority — to collect qualitative, role-specific insights that a survey alone would fail to capture.

5.3 Research Participants

The questionnaire focused on people who frequently utilize public buses or trains. A total of 36 valid responses were gathered; the sample was predominantly composed of university students aged 18–25 (89% of participants), indicative of the accessibility of this demographic online, while the rest consisted of employed, self-employed, and unemployed individuals. At least five participants from a minimum of two other stakeholder categories were incorporated through the follow-up interviews (Section 6), to introduce the viewpoints of transport personnel and authorities not covered by the survey.

5.4 Data Collection Process

The survey was conducted online via Google Forms and gathered 36 responses from frequent public transport users from August 4 to 7, 2026. Participants voluntarily filled out the survey after reviewing a short overview of the study. The responses were examined to uncover shared themes, user requirements, and challenges that inform the design of the RideTrack mobile app. This survey focused solely on primary stakeholders (passengers); additional interviews were held to collect information from secondary and tertiary stakeholders (refer to Section 6).

6 Interview Script

6.1 Interview Script for Persona 1: Daily Bus/Train Commuter (Primary Stakeholder)

Background description of the user	Questions asked from the user	Answers given by the user
This participant is a daily public transport commuter who regularly travels by bus to work. The participant depends on public transportation for reaching the workplace on time and uses a smartphone throughout the day. Uncertainty about vehicle arrival times, unexpected delays, and difficulty obtaining timely transport information can affect the participant's daily schedule. The participant represents a primary RideTrack user who would benefit from real-time vehicle tracking, arrival information, digital ticketing, and service notifications.	• Can you describe your typical daily journey using public transport?	•. I usually travel by bus to work every day. I leave home in the morning and return by bus in the evening.
	• How do you currently find the bus or train you need and check its arrival time?	• I usually know my regular route, but I check Google Maps or ask someone when I need another route. For arrival times, I normally check the scheduled time or wait at the bus stop.
	• What is the biggest problem you experience when travelling by public transport?	• The biggest problem is not knowing exactly when the bus will arrive. Sometimes I have to wait for a long time because of delays or traffic.
	• Have you ever been late or missed a bus/train because of unexpected delays or lack of information?	• Yes. Sometimes the bus arrives later than expected, and I have been late to work because I did not know about the delay.
	• Would knowing the real-time location and estimated arrival time of your bus/train be useful to you? Why?	• Yes. It would help me know when to leave home and reduce unnecessary waiting at the bus stop.
	What information would you want to see when tracking a bus or train?	I would like to see the current location, estimated arrival time, route, destination, and upcoming stops.
	How do you currently purchase your bus or train ticket, and what problems do you face?	I usually buy a ticket from the conductor or at the station. During busy times, there can be queues and it takes extra time.
	Would you use a digital ticket through a mobile application? Why or why not?	Yes. It would be convenient because I could purchase the

		ticket before reaching the station and avoid waiting in a queue.
	If you could have one feature in a public transport application, what would be most useful to you?	If you could have one feature in a public transport application, what would be most useful to you?

6.2 Interview Script for Persona 2: Bus Conductor / Train Ticket Inspector (Secondary Stakeholder)

Background description of the user	Questions asked from the user	Answers given by the user
This participant is a bus conductor with over eight years of experience working on a busy commuter route. He is responsible for collecting fares, managing boarding, and keeping track of how many passengers are on board, and represents a secondary RideTrack stakeholder who would use the application's ticket-validation tools during every trip.	Can you briefly describe your daily responsibilities?	• tickets, help manage boarding and alighting, and try to keep track of how full the bus is.
	What challenges do you face when collecting cash fares?	• Making change takes time, especially during rush hour, and it slows down boarding when the bus is crowded.
	Have you dealt with fare disputes or fraud?	• Occasionally. Some passengers claim they already paid, and without a proper record it's hard to verify.
	How do you currently track how many passengers are on the bus?	• Mostly by counting manually or estimating, which isn't very accurate during busy hours
	How do you think a digital ticketing app could help you?	• If I could scan a QR ticket instead of handling cash, it would be faster to board passengers and I'd have a clear record of who paid.

	What features would make ticket validation easier for you?	• simple scanner that instantly shows if a ticket is valid, plus a running count of passengers on board.

6.3 Interview Script for Persona 3: Transport Authority Operations Officer (Tertiary Stakeholder)

Background description of the officer	Questions asked from the officer	Answers given by the officer
This participant is an Operations Officer at the National Transport Commission (NTC), responsible for monitoring route permit compliance for private and state-owned bus operators and coordinating with Sri Lanka Railways on service performance. The participant represents a regulatory stakeholder who currently depends on manual inspection reports, operator self-declarations, and fragmented phone or email communication to track vehicle performance and handle passenger complaints. The lack of a centralized, real-time system makes it difficult to monitor compliance, respond quickly to disruptions, and generate reports for the Commission. This participant would benefit from a centralized platform offering real-time vehicle tracking, performance dashboards, automated compliance reporting, and the ability to send alerts or notifications directly to operators.	• Can you briefly describe your role at the National Transport Commission (NTC) / Sri Lanka Railways?	I work as an Operations Officer at the NTC. My responsibilities include monitoring route permit compliance for private and state-owned bus operators, overseeing GPS fleet tracking data, and coordinating with Sri Lanka Railways on service performance.
	• How do you currently monitor whether buses and trains are operating on schedule and on their licensed routes?	Most of our monitoring is manual. We rely on periodic inspection reports, operator self-declarations, and complaints from passengers. Some buses have GPS units, but the data isn't centralized, so we can't see everything in real time.
	• What tools or systems do you currently use to track vehicle performance or service performance?	We use a basic GPS monitoring system for a limited number of state-owned buses, spreadsheets for compliance records, and phone or email reports from depot managers. There is no single system that covers all operators.
	• What are the biggest challenges you face in overseeing public transport operations?	The biggest challenges are the lack of real-time, standardized data from all operators, delays in receiving passenger complaints, and the manual effort required to compile compliance reports for the Commission.
	• How do passenger complaints about delays or service issues	Complaints usually come through phone calls, letters, or our call

Background description of the officer	Questions asked from the officer	Answers given by the officer
	currently reach your office, and how long does it take to act on them?	centre, and then get forwarded to the relevant department. It can take several days before a complaint is reviewed and action is taken.
	• Would a centralized digital platform showing real-time bus and train locations across all operators be useful to you? Why?	Yes, definitely. It would let us monitor service levels and route compliance immediately, instead of waiting for reports. It would also help us respond faster to disruptions and identify operators who are not following their schedules.
	• What kind of information or reports would you want to see from such a platform?	I would want real-time vehicle locations, on-time performance statistics, route deviation alerts, ticketing/revenue summaries, and a log of passenger complaints linked to specific vehicles or routes.
	• What concerns, if any, would you have about adopting a system like this across all bus and train operators?	My main concerns are data accuracy and security, ensuring private operators actually install and maintain GPS devices, and making sure the system integrates with our existing NTC records without creating extra administrative work.
	• Would you want the ability to send alerts or notifications to operators through the platform, for example about delays or non-compliance?	Yes, that would be very useful. If we could flag an issue and notify the operator directly through the system, it would speed up corrective action significantly.
	• If you could have one feature in this platform that would make your job easier, what would it be?	A single dashboard that shows the real-time status, compliance history, and performance rating of every registered bus and train service in one place, so I don't have to consolidate data from multiple sources manually.

7 Personas and Empathy Maps

7.1 Daily Bus/Train Commuter (Primary Stakeholder) - IT23665620

PRIMARY PERSONA

DAILY BUS/TRAIN COMMUTER



SNAPSHOT

Name

Kasun Perera

Age

28

Gender

Male

Location

Colombo, Sri Lanka

Occupation

Office Employee

Background

Kasun is a daily office commuter who relies on public buses to travel between home and work. He follows his regular route most days and uses his smartphone to check transport information when needed. He usually depends on scheduled arrival times or Google Maps, but unexpected traffic and delays make it difficult to know exactly when his bus will arrive. This can lead to unnecessary waiting and sometimes causes him to be late for work.

PERSONAL

Characteristics

- Responsible and punctual
- Regular public transport user
- Smartphone user
- Prefers simple and quick applications
- Values accurate information

Hobbies and Interests

- Watching YouTube
- Listening to music
- Spending time with family
- Using mobile applications
- Browsing the internet
- Travelling around the city

GOALS

Primary goals

- Know the exact or estimated bus arrival time
- Track the bus in real time
- Know the current location of the bus
- Avoid missing the bus
- Avoid unnecessary waiting
- Reach work on time
- Find the correct route easily

Motivations

- Save travel time
- Reach work on time
- Reduce waiting at bus stops
- Make daily commuting less stressful
- Plan journeys more effectively
- Have reliable transport information

CHALLENGES

Main pain points

- Uncertain bus arrival times
- Unexpected delays
- Long waiting times
- Difficulty planning journeys
- Queues for physical tickets
- Lack of real-time information

Frustrations

- Waiting without knowing when the bus will arrive.
- Finding out about delays too late.
- Not knowing the current location of the bus.
- Being unable to accurately plan when to leave home.

EMPATHY MAP

Primary Stakeholder – Daily Bus/Train Commuter

Occupation: Office Employee | Age: 28 | Gender: Male | Location: Colombo, Sri Lanka

SAYS

- "I never know exactly when my bus will arrive."
- "Sometimes I wait for a long time because of delays."
- "I wish I could check the bus before leaving home."
- "It's frustrating when I miss the bus and have to wait for the next one."
- "Buying tickets at the station takes too much time."

THINKS

- Will the bus arrive on time today?
- Should I leave home now or wait a little longer?
- What if the bus is delayed again?
- Is there a better route or bus I can take?
- How can I avoid long queues for tickets?



DOES

- Checks bus routes and schedules
- Uses Google Maps for directions
- Waits at the bus stop
- Checks time repeatedly
- Asks others about bus arrival
- Buys ticket from conductor or station
- Adjusts plans when buses are delayed
- Uses smartphone during the journey

PAINS

- Uncertain arrival times
- Unexpected delays
- Long waiting times
- Risk of being late to work
- Limited delay information
- Missing buses
- Queues for physical tickets
- Difficulty planning journeys

FEELS

- Frustrated when the bus is late
- Worried about being late to work
- Impatient while waiting
- Confused when there is no clear information
- Relieved when the bus arrives
- Happy when the journey is smooth
- Stressed during unexpected delays
- Satisfied when information is accurate

GAINS

- Real-time bus location
- Accurate estimated arrival times
- Delay and service notifications
- Easy route search
- Information about upcoming stops
- Digital ticket purchasing
- Avoid long ticket queues
- Plan journey before leaving home
- Reach work on time
- All transport info in one app

7.2 Bus Conductor / Train Ticket Inspector (Secondary Stakeholder) - IT23634626

SECONDARY PERSONA

BUS CONDUCTOR/ TRAIN TICKET INSPECTOR



SNAPSHOT

Name

Dinuka Bandara

Age

30

Gender

male

Location

Colombo, Sri Lanka

Occupation

Bus Conductor

Background

Dinuka has worked as a bus conductor on a busy Colombo commuter route for over eight years. He is responsible for collecting fares, managing boarding during rush hour, and keeping an informal count of passengers on board. Handling cash fares and change during peak hours slows down boarding and occasionally leads to disputes over payment, so he is interested in tools that can speed up ticket validation and give him a clearer, more reliable record of who has paid.

PERSONAL

Characteristics

- Experienced and efficient under pressure
- Comfortable using a basic smartphone
- Works long shifts on a fixed route
- Values speed and simplicity over extra features

Hobbies and Interests

- Watching cricket
- Spending time with family on days off
- Chatting with regular passengers

GOALS

Primary goals

- Validate tickets quickly during boarding.
- Keep an accurate count of passengers on board
- Reduce disputes over fare payment
- Spend less time handling cash and change

Motivations

- Keep the bus moving on schedule
- Avoid conflict with passengers over payment
- Make his job less physically and mentally tiring
- Support a smoother experience for passengers

CHALLENGES

Main pain points

- Making change slows down boarding at busy stops
- No reliable record to resolve fare disputes
- Manual passenger counting is inaccurate
- Handling cash all day is inconvenient and risky

Frustrations

- Delays caused by slow cash transactions
- Passengers disputing whether they already paid
- Not knowing exactly how full the bus is
- Extra pressure during rush-hour boarding

EMPATHY MAP
Dinuka Bandara – Bus Conductor

Secondary Empathy Map – Bus Conductor / Train Ticket Inspector

Occupation: Bus Conductor | Age: 30 | Gender: Male | Location: Colombo, Sri Lanka

SAYS
- "Making change takes too long when the bus is full."
- "I need a fast way to check if a ticket is valid."
- "Sometimes passengers argue that they already paid."
- "It would help if passengers could check arrival times themselves instead of asking me."



THINKS
- I need to keep the bus on schedule no matter what.
- A quick QR scan would save so much time during rush hour.
- It would help to know exactly how many people are on board.
- I worry the device might run out of battery or lose signal at the wrong moment.

DOES
- Collects cash fares and issues paper tickets
- Manages boarding and alighting at every stop
- Estimates passenger count by eye
- Handles disputes over payment
- Answers passengers' questions about arrival times and delays

Age: 30
Gender: Male
Location: Colombo, Sri Lanka
Occupation: Bus Conductor

FEELS
- Stressed during peak-hour boarding
- Frustrated by repeated fare disputes
- Satisfied when boarding goes smoothly
- Open to tools that make his job easier
- Anxious about technical failures (battery, signal) during a busy trip

PAINS
- Making change slows down boarding at busy stops
- No reliable record to resolve fare disputes
- Manual passenger counting is inaccurate
- Handling cash all day is inconvenient and risky
- No fallback plan if the digital device fails mid-route

GAINS
- Faster boarding through instant QR ticket validation
- Clear, disputable-proof digital record of every payment
- Automatic, accurate passenger occupancy count
- Less cash handling — reduced risk and stress
- More time to focus on safe driving/route management instead of manual counting

7.3 Transport Authority Operations Officer (Tertiary Stakeholder) – IT23635302

TERTIARY PERSONA

TRANSPORT AUTHORITY OPERATIONS OFFICER



SNAPSHOT

Name

Amila Jayasinghe

Age
38

Gender
Male

Location
Colombo, Sri Lanka

Occupation
Operations Officer, NTC

Background

Amila is an Operations Officer at the National Transport Commission, overseeing route-permit compliance and service performance for both bus and train operators. He depends on scattered manual reports, occasional GPS data, and passenger complaints to monitor the public transport network. The lack of a centralized system makes it difficult for him to respond quickly to service issues or hold non-compliant operators accountable.

PERSONAL

Characteristics

- Detail-oriented and process-driven
- Responsible for regulatory compliance
- Moderate technology user, prefers clear dashboards over raw data
- Focused on accountability and public service quality

Hobbies and Interests

- Reading transport policy research
- Cycling on weekends
- Volunteering in community road-safety programs

GOALS

Primary goals

- Monitor bus and train locations in real time
- Identify and respond to delays quickly.
- Analyse passenger and occupancy statistics by route
- Improve overall service reliability

Motivations

- Reduce passenger complaints about unreliable service
- Support data-driven route and fleet planning
- Improve public trust in the transport system
- Respond to disruptions before they escalate

CHALLENGES

Main pain points

- No real-time visibility into vehicle locations
- Delays are only discovered after passenger complaints
- Limited data on route-level overcrowding
- Difficult to coordinate a fast response with

Frustrations

- Reacting to problems instead of preventing them
- Lack of a centralized live dashboard
- Difficulty proving service issues without data
- Slow communication with bus and train operators

EMPATHY MAP

Amila Jayasinghe – Operations Officer, NTC

Tertiary Empathy Map – Transport Authority Operations Officer

Occupation: Operations Officer, NTC | Age: 38 | Gender: Male | Location: Colombo, Sri Lanka

SAYS

- "We need real-time, accurate data from every operator, not just a few."
- "Compliance reporting takes far too long to compile manually."
- Passenger complaints reach us days after the issue occurred."
- Data privacy and system security have to be guaranteed."

THINKS

- How can we monitor hundreds of vehicles without more staff?
- Will private operators cooperate with mandatory GPS tracking?
- Is the incoming data reliable enough for regulatory decisions?
- How do we fairly enforce action against non-compliant operators?



Age: 38
Gender: Male
Location: Colombo, Sri Lanka
Occupation: Operations Officer, NTC

DOES

- Reviews operator compliance and route-permit records.
- Monitors GPS fleet dashboards for state-owned vehicles.
- Handles escalated passenger complaints and coordinates depots.
- Conducts periodic inspections and issues route permits.
- Liaises with Sri Lanka Railways on cross-mode service data.

FEELS

- Frustrated by outdated, manual reporting processes.
- Anxious about accountability when services are delayed.
- Hopeful that digital tracking will reduce complaint volume.
- Cautious about relying on data of uncertain accuracy.
- Satisfied when a system genuinely reduces workload.

PAINS

- Fragmented data from multiple, inconsistent operators.
- No standardized real-time monitoring across all services.
- Manual, slow complaint-handling process.
- Limited enforcement tools against non-compliant operators.
- Resistance from private operators to adopt new technology.

GAINS

- Centralized real-time dashboard for all operators.
- Automated compliance alerts and reporting.
- Data-driven decisions on routes and permits.
- Faster response to passenger complaints.
- Streamlined communication with operators and SL Railways.

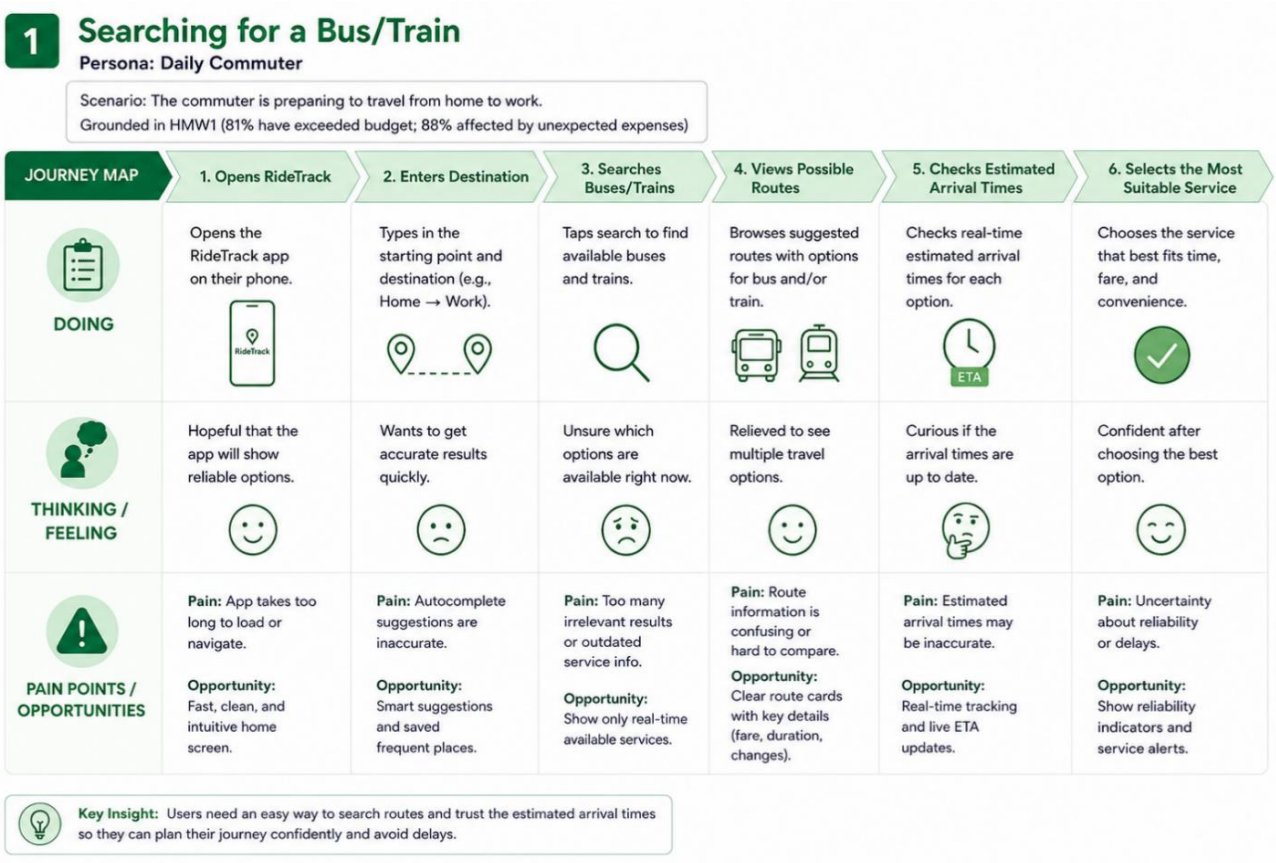
8 User Stories

1. As a **daily bus commuter**, I want to see the live location of my bus so that I know when to leave home.

2. As a **train passenger**, I want to purchase a digital ticket before arriving at the station so that I can avoid waiting in long queues.

3. As a **bus conductor**, I want to scan QR tickets so that I can verify passengers efficiently.
4. As a **train ticket inspector**, I want to view ticket validity instantly so that I can reduce fraud.
5. As a **transport authority officer**, I want to monitor vehicles' locations and delays so that I can improve service quality

9 User Journey Maps



2 Tracking the Vehicle

Persona: Daily Commuter

Scenario: The commuter has selected a bus/train and is waiting for it.
Grounded in HMW1 (81% have exceeded budget; 88% affected by unexpected expenses)

JOURNEY MAP	1. Selects the Vehicle	2. Views its Current Location	3. Checks Estimated Arrival Time	4. Views Upcoming Stops	5. Waits for the Vehicle
DOING	Selects the bus/train from the search results or favorite list. 	Views the vehicle's live location on the map. 	Checks the estimated arrival time at their stop. 	Views the list of upcoming stops and its progress. 	Waits for the vehicle at the stop while tracking updates. 
THINKING / FEELING	Hopeful I picked the right one. 	Glad to see exactly where it is. 	Hopeful it will arrive on time. 	Good to know what stops are next. 	Hope it arrives soon and not delayed. 
PAIN POINTS / OPPORTUNITIES	**Pain:** Not sure if this is the right vehicle or if it's on time. **Opportunity:** Show key details clearly (route, destination, frequency).	**Pain:** Hard to know where the vehicle actually is. **Opportunity:** Live GPS tracking on map with real-time updates.	**Pain:** ETA may be inaccurate or not available. **Opportunity:** Accurate ETA based on live traffic and conditions.	**Pain:** Don't know which stop comes next or if any stops are skipped. **Opportunity:** Show upcoming stops and real-time progress along the route.	**Pain:** Uncertainty causes unnecessary waiting or missed connections. **Opportunity:** Delay notifications and service alerts.

Key Insight: Users need live vehicle tracking, accurate ETA, and timely delay notifications so they can wait with confidence and plan their onward journey better.

3 Purchasing a Digital Ticket

Persona: Daily Commuter

Scenario: The commuter wants to avoid spending additional time buying a physical ticket.
Grounded in HMW1 (81% have exceeded budget; 88% affected by unexpected expenses)

JOURNEY MAP	1. Selects the Journey	2. Selects the Required Ticket	3. Makes the Payment	4. Receives a Digital Ticket	5. Opens the QR Code When Required
DOING	Confirms the route, date, and travel details. 	Chooses the ticket type (single, return, day pass, etc.) and quantity. 	Selects a payment method and completes the transaction. 	Receives a digital ticket instantly in the app. 	Shows the QR code to the scanner when boarding. 
THINKING / FEELING	Happy to plan ahead and save time. 	Wants the right ticket that fits the journey and budget. 	Wants a fast, secure, and hassle-free payment. 	Relieved to have the ticket ready without standing in line. 	Confident and ready to board quickly. 
PAIN POINTS / OPPORTUNITIES	**Pain:** Not sure if the selected service is eligible for digital tickets. **Opportunity:** Clearly show digital ticket availability during search.	**Pain:** Confusing ticket options or unclear pricing. **Opportunity:** Simple ticket categories with clear prices and validity info.	**Pain:** Payment failures or limited payment options. **Opportunity:** Multiple secure payment methods and instant confirmation.	**Pain:** Ticket not received or hard to find in the app. **Opportunity:** Instant delivery with easy access in 'My Tickets' plus email backup.	**Pain:** Poor network or scanner issues at the gate. **Opportunity:** Fast-loading QR code with offline fallback and clear instructions.

Key Insight: Users want a quick, reliable digital ticketing experience that eliminates queues and lets them board with confidence using QR-code validation.

4 Completing the Journey

Persona: Daily Commuter

Scenario: The commuter boards the vehicle and travels to the destination.
Grounded in HMW1 (81% have exceeded budget; 88% affected by unexpected expenses)

JOURNEY MAP	1. Boards the Bus/Train	2. Shows or Scans the Digital Ticket	3. Checks Upcoming Stops if Needed	4. Receives Journey/ Delay Information	5. Gets Off at the Destination
DOING	Boards the bus or train from the stop/station. 	Shows the digital ticket QR code or lets it be scanned by the inspector/driver. 	Checks the list of upcoming stops and the live progress of the journey if needed. 	Receives real-time updates about delays, reroutes, or platform/stop changes. 	Gets off the vehicle at the destination stop or station. 
THINKING / FEELING	Relieved to be on board and on the right service.	Confident and ready with a valid ticket.	Wants to know when to get ready to get off.	Appreciates being informed so they can plan ahead.	Happy to have reached destination on time.
PAIN POINTS / OPPORTUNITIES	**Pain:** Crowded vehicles or unclear boarding information. **Opportunity:** Show real-time crowding level and platform/ boarding guidance.	**Pain:** Poor network, slow scans, or ticket not recognized. **Opportunity:** Offline QR code support and fast, reliable validation.	**Pain:** Unclear stop announcements or not knowing when to get off. **Opportunity:** Clear upcoming stop list with live progress and alerts before your stop.	**Pain:** Sudden delays or route changes without any notice. **Opportunity:** Real-time delay alerts, reason for delay, and updated arrival estimates.	**Pain:** Missing the stop or not knowing the exact alighting point. **Opportunity:** Arrival alerts and "next stop is your destination" notifications.

Key Insight: Users need continuous, reliable information throughout the ride—upcoming stops, real-time updates, and timely alerts—so they can travel with confidence and arrive on time.

2 Validating Digital Tickets During a Journey

Persona: Bus Conductor / Train Ticket Inspector

Scenario: Scanning and verifying passenger tickets while managing boarding at a busy stop.
Grounded in HMW2 (ticket purchase/validation is slow and cash-dependent)

JOURNEY MAP	1. Bus Arrives at Stop	2. Scans QR Ticket	3. Verifies Validity	4. Handles an Exception	5. Reviews Passenger Count
DOING	Opens the boarding doors as passengers begin to line up. 	Scans each passenger's digital ticket using the app. 	Confirms that the ticket is valid for the route and travel time. 	Assists a passenger whose ticket fails to scan. 	Checks the running passenger count for the trip. 
THINKING / FEELING	Focused on keeping boarding fast and organised.	Wants an instant and unambiguous scan result.	Confident when the system automatically confirms validity.	Wants a quick and simple way to resolve the issue without conflict.	Reassured when the passenger count is accurate and up to date.
PAIN POINTS / OPPORTUNITIES	**Pain:** Crowded stops make it difficult to process passengers quickly. **Opportunity:** Enable fast, one-tap scanning with instant feedback.	**Pain:** Slow or unclear scan results can delay boarding. **Opportunity:** Use clear green/red visual feedback with a sound cue.	**Pain:** It can be difficult to manually determine whether a ticket is genuine and valid. **Opportunity:** Automatically flag invalid, expired, or already-used tickets.	**Pain:** There is no easy fallback when a QR ticket cannot be scanned. **Opportunity:** Provide manual lookup using phone number or booking ID.	**Pain:** Manual passenger counting can be inaccurate. **Opportunity:** Automatically update occupancy/passenger count with every successful scan.

10 How Might We (HMW) Statements

The following HMW statements are each grounded in a distinct pain point identified from the 40 survey responses (Section 6) and the stakeholder interviews (Section 6).

HMW 1

Pain point: Passengers don't know when their bus or train will actually arrive.

Evidence: 78% have missed transport or arrived late due to unclear arrival times; ~55% currently track buses/trains by guessing or asking others.

How might we help passengers know exactly when their bus or train will arrive?

HMW 2

Pain point: Buying and validating a ticket is slow and still mostly cash-based.

Evidence: Only 35% of respondents consistently use a digital ticket; conductor interview highlights slow cash handling and unclear fare disputes.

How might we make purchasing and validating tickets faster and easier?

HMW 3

Pain point: Transport authorities have no real-time visibility into the network.

Evidence: Tertiary stakeholder interview: delays are discovered only through passenger complaints, with no live vehicle data available.

How might we help transport authorities reduce delays and improve route management?

HMW 4

Pain point: Passengers are never notified when a service is delayed or cancelled.

Evidence: 85% experienced a delay or cancellation with no notification at all.

How might we notify passengers immediately about service disruptions?

HMW 5

Pain point: Buses and trains are frequently overcrowded with no way to know in advance.

Evidence: 82% experienced an overcrowded bus/train; three unprompted requests for an occupancy indicator before boarding.

How might we reduce overcrowding by showing vehicle occupancy?

11 Functional Requirements

Functional requirements describe the features and services that the real-time vehicle tracking mobile application must provide to its users.

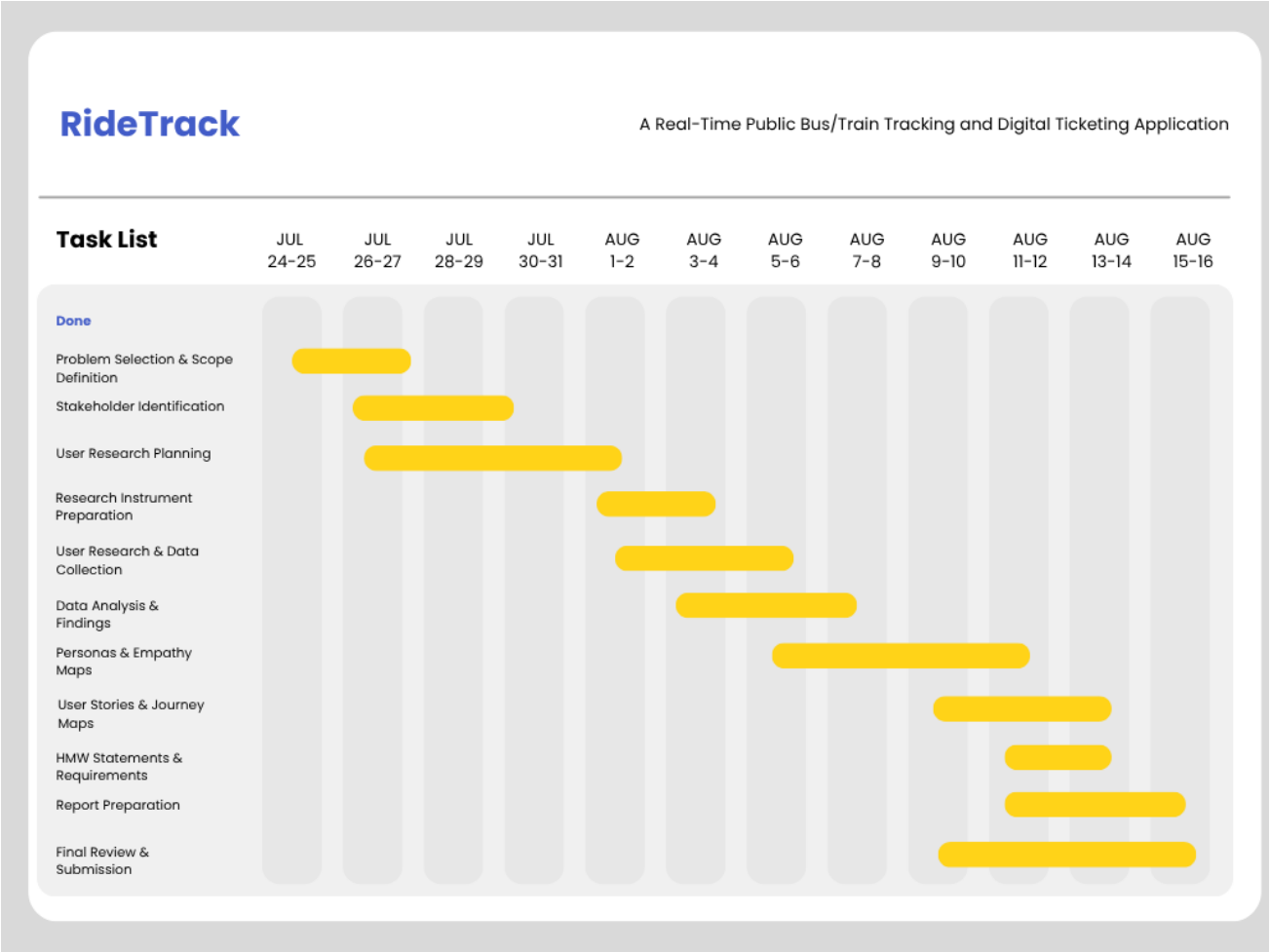
ID	Functional Requirement
FR1	The system shall allow users to register a new account using their email address or mobile number.
FR2	The system shall allow registered users to log in securely.
FR3	The system shall allow users to search for bus/train routes and view upcoming arrivals.
FR4	The system shall display the live location of buses/trains on a map with an estimated arrival time.
FR5	The system shall allow users to purchase a digital QR-code ticket for a selected route.
FR6	The system shall allow conductors/inspectors to scan and validate a passenger's QR ticket.
FR7	The system shall track and display the current passenger occupancy of a vehicle where available.
FR8	The system should notify passengers of delays, cancellations, or route changes in real time.
FR9	The system shall provide an operations dashboard showing live vehicle locations and delay alerts for transport authority users.
FR10	The system shall generate reports and statistics on route performance, delays, and occupancy.
FR11	The system shall maintain a transaction/trip history for each passenger.
FR12	The system shall allow users to update their profile information and application preferences.

12 Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the real-time vehicle tracking mobile application.

ID	Non-Functional Requirement
NFR1	The application shall provide a simple, intuitive, and user-friendly interface that is easy to navigate for all three user types.
NFR2	The application shall update vehicle location and ETA information within 5 seconds under normal network conditions.
NFR3	User data and digital tickets shall be protected using secure authentication and encrypted communication.
NFR4	The application shall be available whenever users require access, except during scheduled maintenance.
NFR5	The application shall maintain the confidentiality of users' personal and payment information.
NFR6	The application shall support Android devices running recent operating system versions.
NFR7	The application should provide clear error messages and prevent accidental loss of ticket or payment data.
NFR8	The application should use readable fonts, accessible colours, and simple navigation to improve usability and accessibility.
NFR9	The application shall be scalable to support additional routes, operators, and features such as fare-card integration in future releases.
NFR10	The application shall cache a purchased ticket locally so it remains scannable even with a weak or lost network connection.

13 Gantt Chart



14 References

- Gray, D., Brown, S. and Macanufo, J. (2010) Gamestorming: A Playbook for Innovators, Rulebreakers, and Changemakers. Sebastopol: O'Reilly Media. (Empathy Mapping method)
- IDEO.org (2015) The Field Guide to Human-Centered Design. San Francisco: IDEO.org. (How Might We Statement)
- Norman, D. A. (2013) The Design of Everyday Things. Revised and expanded edition. New York: Basic Books.
- Nielsen Norman Group (n.d.) Personas and User Journey Mapping. Available at: https://www.nngroup.com/ (Accessed: 2026).
- Cohn, M. (2004) User Stories Applied: For Agile Software Development. Boston: Addison-Wesley.
- Google LLC (2026) Google Forms [Online survey tool].

15. Appendix

Google Form Questionnaire –

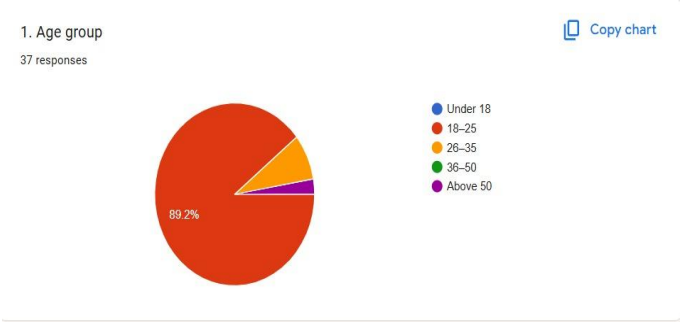
https://docs.google.com/forms/d/e/1FAIpQLSeoQh-8Q7r_wCgOJCuHzJ2Yzz65FVPsTx4n5jCu807wQ89Crg/viewform?usp=publish-editor

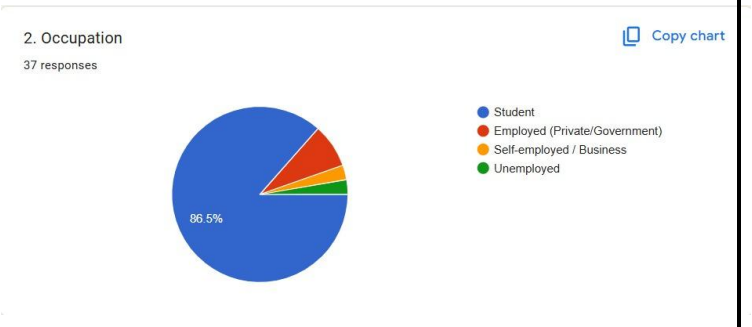
Google Form

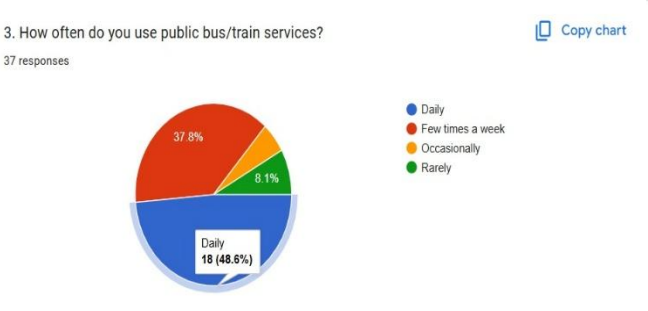
User Research Questionnaire for a Real-Time Bus and Train Tracking & Digital Ticketing app to make public transportation form link:
https://docs.google.com/forms/d/e/1FAIpQLSeoQh-8Q7r_wCgOJCuHzJ2Yzz65FVPsTx4n5jCu807wQ89Crg/viewform?usp=publish-editor

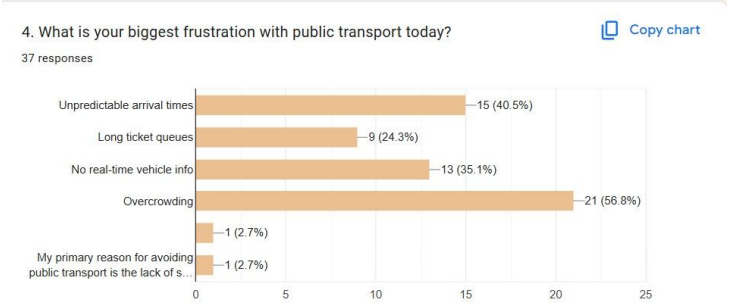
Participant responses Summary

The questionnaire was distributed using Google Forms to individuals who had Real-Time Bus and Train Tracking & Digital Ticketing app





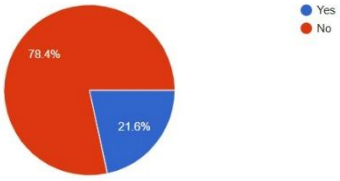




5. Do you currently use any app to check bus/train schedules or locations?

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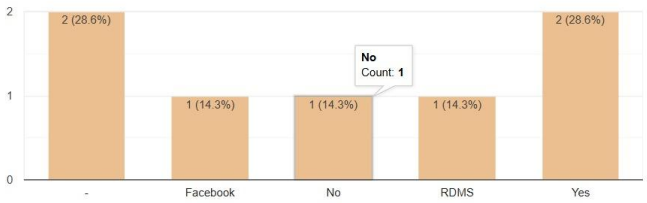
37 responses



6. If yes, Which one?

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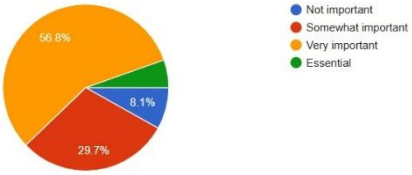
7 responses



8. How important is real-time vehicle tracking (seeing exactly where your bus/train is) to you?

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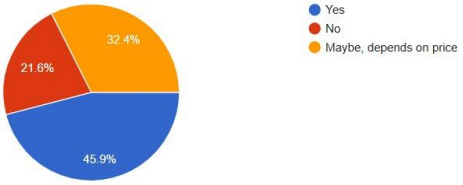
37 responses



9. Would you prefer to buy tickets digitally (in-app) instead of cash/counter?

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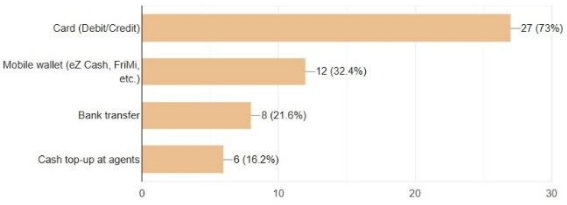
37 responses



10. What payment methods would you want supported for digital tickets?

Copy chart

37 responses



11. How much advance notice do you want before your bus/train arrives (via notification)?

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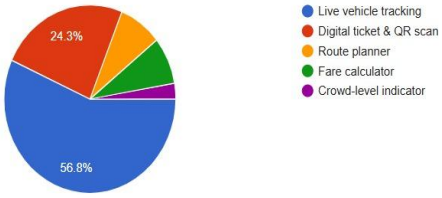
37 responses



12. Which single feature matters most to you? (pick one)

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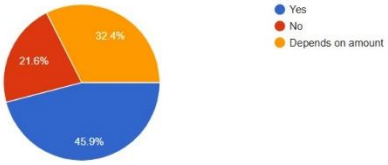
37 responses



13. Would you pay a small convenience fee for guaranteed seat reservation or skip-the-queue ticketing?

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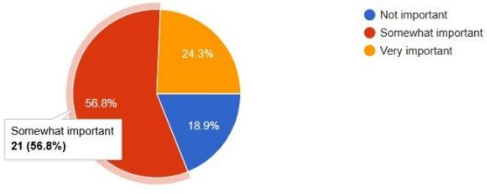
37 responses



14. How important is offline access (e.g., viewing your ticket without internet)?

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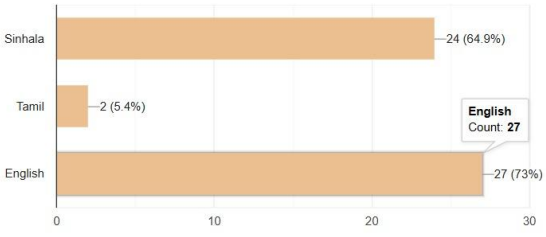
37 responses



15. What language would you prefer for the app interface?

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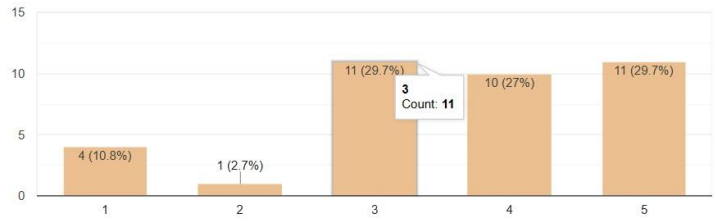
37 responses



16. On a scale of 1–5, how likely are you to use this app regularly if launched?

Copy chart

37 responses



17. Any other suggestions or features you'd like to see in this app?

11 responses

No

Yes

Yes

I like this features

Seats availability tracking

-

This is very good decision & very importance for daily bus/train users.

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